

# Randomization Forensics Audit

## Table of contents

|          |                                                                |           |
|----------|----------------------------------------------------------------|-----------|
| <b>1</b> | <b>Scope</b>                                                   | <b>1</b>  |
| <b>2</b> | <b>Load Inputs</b>                                             | <b>2</b>  |
| <b>3</b> | <b>Trial Metadata Snapshot</b>                                 | <b>3</b>  |
| <b>4</b> | <b>Pooled Summary Metrics</b>                                  | <b>6</b>  |
| 4.1      | Interpretation Key (Heathers-aligned) . . . . .                | 6         |
| <b>5</b> | <b>Row-Level Signals</b>                                       | <b>7</b>  |
| 5.1      | Lowest recalculated p-values . . . . .                         | 7         |
| 5.2      | Largest reported vs recalculated p-value gaps . . . . .        | 7         |
| <b>6</b> | <b><code>simdistr</code> Variable-Level Signals</b>            | <b>8</b>  |
| <b>7</b> | <b>Diagnostics</b>                                             | <b>9</b>  |
| 7.1      | Distribution of recalculated row p-values . . . . .            | 9         |
| 7.2      | Reported vs recalculated row p-values . . . . .                | 10        |
| 7.3      | Lowest <code>simdistr</code> variable-level p-values . . . . . | 11        |
| <b>8</b> | <b>Interpretation Notes</b>                                    | <b>11</b> |

## 1 Scope

This report summarizes randomization-forensics signals from baseline characteristics for the configured study.

Inputs are generated by:

- `scripts/extract_randomization_table1.py`

- scripts/build\_randomization\_inputs.py
- scripts/run\_randomization\_forensics.R

## 2 Load Inputs

```

repo_root <- normalizePath("../", winslash = "/", mustWork = TRUE)
study_id <- Sys.getenv("FORENSICS_STUDY_ID", "lungtime")
study_label <- Sys.getenv("FORENSICS_STUDY_TITLE", study_id)
input_dir <- file.path(repo_root, "reports", "randomization", study_id)

required_files <- c(
  "row_level_results.csv",
  "pooled_pvalues.csv",
  "simdistr_variable_pvalues.csv",
  "simdistr_combined_pvalues.csv"
)
missing_files <- required_files[!file.exists(file.path(input_dir, required_files))]
if (length(missing_files) > 0) {
  stop(
    "Missing expected analysis outputs: ",
    paste(missing_files, collapse = ", "),
    ". Run scripts/run_randomization_forensics.R first."
  )
}

row_level <- read.csv(file.path(input_dir, "row_level_results.csv"), check.names = FALSE)
pooled <- read.csv(file.path(input_dir, "pooled_pvalues.csv"), check.names = FALSE)
sim_var <- read.csv(file.path(input_dir, "simdistr_variable_pvalues.csv"), check.names = FALSE)
sim_combined <- read.csv(file.path(input_dir, "simdistr_combined_pvalues.csv"), check.names = FALSE)

ascii_text <- function(x) {
  x <- gsub(" ", ">=", x, fixed = TRUE)
  x <- gsub("-", "-", x, fixed = TRUE)
  x
}

row_level$variable <- ascii_text(row_level$variable)
row_level$level <- ascii_text(row_level$level)
sim_var$variable <- ascii_text(sim_var$variable)
sim_var$level <- ascii_text(sim_var$level)

```

### 3 Trial Metadata Snapshot

```
meta_path <- file.path(repo_root, "data", "processed", "randomization", study_id, "randomization_metadata.csv")
if (file.exists(meta_path)) {
  metadata <- read.csv(meta_path, check.names = FALSE)
  knitr::kable(metadata)
} else {
  cat("Metadata file not found at:", meta_path)
}
```



| al-                               | allo-         | ca-      | strat-                                                 | source | page | proce- |
|-----------------------------------|---------------|----------|--------------------------------------------------------|--------|------|--------|
| lo-ran-                           | ca-           | tion     | conf-                                                  | source | page | proce- |
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| trialbell_n_a                     | tion          | method   | tion                                                   | source | page | proce- |
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| proptoush272228-27253260025004333 |               |          |                                                        | 1      | 3    | 1      |
| tients                            | tients        | will     | thelancet.com/respiratory on March 25, 2026            |        |      |        |
| aged                              | aged          | use      | See Online/Comment https://doi.org/10.1016/            |        |      |        |
| 16                                | 16            | min-     | S2213-2600(25)00471-0 <i>Joint last authors</i>        |        |      |        |
| years                             | years         | imi-     | <i>Members listed in the appendix (pp 4–7)</i>         |        |      |        |
| and                               | and           | sa-      | <i>Tropical and Infectious Disease Unit, Liverpool</i> |        |      |        |
| older                             | older         | tion     | <i>University Hospitals NHS Foundation Trust,</i>      |        |      |        |
| with                              | with          | with     | <i>Liverpool, UK (S Todd PhD FRCP, S Aston</i>         |        |      |        |
| sus-                              | sus-          | NEWS2    | <i>PhD FRCP, Prof N French PhD FRCP);</i>              |        |      |        |
| pected                            | pected        | score    | <i>Centre for Trials Research, College of</i>          |        |      |        |
| sep-                              | sep-          | ( or     | <i>Biomedical and Life Sciences, Cardiff</i>           |        |      |        |
| sis                               | sis           | <        | <i>University, Cardiff, UK (J Euden PhD, J</i>         |        |      |        |
| were                              | were          | 5)       | <i>Condie MSc, L Brookes-Howell PhD, S Gilbert</i>     |        |      |        |
| ran-                              | ran-          | and      | <i>BSc, W Maboshe PhD, S Milosevic PhD, A</i>          |        |      |        |
| domly                             | domly         | site     | <i>Thomas PhD, E Thomas-Jones PhD, P</i>               |        |      |        |
| as-                               | as-           | as       | <i>Pallmann PhD); Institute of Systems,</i>            |        |      |        |
| signed                            | signed        | bal-     | <i>Molecular and Integrative Biology, University</i>   |        |      |        |
| to ei-                            | to ei-        | anc-     | <i>of Liverpool, Liverpool, UK (S Aston); Hull</i>     |        |      |        |
| ther                              | ther          | ing      | <i>York Medical School, University of York, UK</i>     |        |      |        |
| usual                             | usual         | fac-     | <i>(Prof G Barlow MD); Hull University Teaching</i>    |        |      |        |
| care                              | care          | tors     | <i>Hospitals NHS Trust, Hull, UK (Prof G</i>           |        |      |        |
| or                                | or            | and      | <i>Barlow); UK Sepsis Trust, Walsall, UK (J</i>        |        |      |        |
| procalcitonin                     | procalcitonin | Carman); | <i>Institute of Infection, Veterinary and</i>          |        |      |        |
| guided                            | guided        | a        | <i>Ecological Sciences, University of Liverpool,</i>   |        |      |        |
| care                              | care          | ran-     | <i>Liverpool, UK (Prof E D Carrol MD FRCPCH,</i>       |        |      |        |
| in a                              | in a          | dom      | <i>Prof N French); NHS England and NHS</i>             |        |      |        |
| 1:1                               | 1:1           | ele-     | <i>Improvement North- East and Yorkshire, Leeds,</i>   |        |      |        |
| ratio                             | ratio         | ment     | <i>UK Procalcitonin testing combined with</i>          |        |      |        |
| via a                             | via a         | to       | <i>NEWS2 evaluation compared with usual care</i>       |        |      |        |
| cen-                              | cen-          | re-      | <i>based on NEWS2 for identification of sepsis</i>     |        |      |        |
| trally                            | trally        | duce     | <i>and antibiotic initiation in the emergency</i>      |        |      |        |
| con-                              | con-          | the      | <i>department in England and Wales (PRONTO):</i>       |        |      |        |
| trolled                           | trolled       | risk     | <i>a multicentre, randomised, controlled,</i>          |        |      |        |
| web-                              | web-          | of       | <i>open-label, phase 3 trial Stacy Todd, Joanne</i>    |        |      |        |
| based                             | based         | sub-     | <i>Euden, Jennifer Condie, Stephen Aston, Gavin</i>    |        |      |        |
| ran-                              | ran-          | ver-     | <i>Barlow, Lucy Brookes-Howell, Julie Carman,</i>      |        |      |        |
| domi-                             | domi-         | sion.    | <i>Enitan D Carrol, Stephanie Gilbert, Philip</i>      |        |      |        |
| sa-                               | sa-           |          | <i>Howard, Kerenza Hood, Matthew Inada-Kim,</i>        |        |      |        |
| tion                              | tion          |          | <i>Martin J Llewelyn, Wakunyambo Maboshe,</i>          |        |      |        |
| pro-                              | pro-          |          | <i>Fiona McGill, Sarah Milosevic, Emmanuel</i>         |        |      |        |
| gramme                            | gramme.       |          | <i>Nsutebu, Paul Schmidt, Andrew Tabner, David</i>     |        |      |        |
|                                   |               |          | <i>Taylor-Robinson, Abin Thomas, Emma</i>              |        |      |        |

## 4 Pooled Summary Metrics

### 4.1 Interpretation Key (Heathers-aligned)

The metrics below are randomization-forensics checks on baseline Table 1 distributions, aligned with the screening logic discussed in Heathers' overview of STALT/CSF-style methods:

- `n_rows_reported`: number of baseline rows with manuscript-reported p-values.
- `n_rows_recalc`: number of baseline rows with recalculated row-level p-values.
- `stouffer_reported`: one-sided Stouffer combination of reported row p-values.
- `stouffer_recalc`: one-sided Stouffer combination of recalculated row p-values.
- `fisher_reported`: Fisher combination of reported row p-values.
- `fisher_recalc`: Fisher combination of recalculated row p-values.
- `simdistr_combined_pvalue`: simulation-based combined p-value from `simdistr`.

Interpretation rule of thumb:

- Smaller combined p-values indicate less compatibility with expected random baseline balance.
- Larger combined p-values indicate no low-tail signal for imbalance under these checks.
- No single metric is dispositive; methods are screening tools and should be interpreted with design/context checks from protocol and data provenance.

```
knitr::kable(pooled, digits = 6)
```

| trial_id                          | n_rows_reported | n_rows_recalc | stouffer_reported | stouffer_recalc | fisher_reported | fisher_recalc |
|-----------------------------------|-----------------|---------------|-------------------|-----------------|-----------------|---------------|
| pronto_lrm_2026_s2213260025004333 | 43              | NA            | 0.952548          | NA              | 0.900389        |               |

```
knitr::kable(sim_combined, digits = 6)
```

| simdistr_combined_pvalue | trial_name                        |
|--------------------------|-----------------------------------|
| 0.371086                 | pronto_lrm_2026_s2213260025004333 |

```
n_rows <- nrow(row_level)
n_flagged_delta <- sum(row_level$flagged_p_delta_0_05, na.rm = TRUE)
cat("Rows analyzed:", n_rows, "\n")
```

Rows analyzed: 43

```
cat("Rows with |reported p - recalculated p| >= 0.05:", n_flagged_delta, "\n")
```

Rows with |reported p - recalculated p| >= 0.05: 0

## 5 Row-Level Signals

### 5.1 Lowest recalculated p-values

```
row_by_p <- row_level[order(row_level$row_chisq_p, na.last = TRUE), ]
knitr::kable(head(row_by_p[, c("variable", "level", "row_chisq_p", "reported_p")], 10), digits = 5)
```

|    | variable                       | level                   | row_chisq_p | reported_p |
|----|--------------------------------|-------------------------|-------------|------------|
| 22 | Ethnicity                      | Arab                    | 0.02590     | NA         |
| 11 | Ethnicity                      | Asian                   | 0.07283     | NA         |
| 31 | Comorbidities                  | Heart Failure           | 0.08453     | NA         |
| 19 | Ethnicity                      | Caribbean               | 0.18259     | NA         |
| 28 | Number of Listed Comorbidities | Missing                 | 0.19447     | NA         |
| 33 | Comorbidities                  | Ischaemic Heart Disease | 0.20153     | NA         |
| 29 | Comorbidities                  | Diabetes Mellitus       | 0.28549     | NA         |
| 25 | Number of Listed Comorbidities | 0                       | 0.29591     | NA         |
| 12 | Ethnicity                      | Indian                  | 0.29840     | NA         |
| 3  | Ethnicity                      | White Irish             | 0.30610     | NA         |

### 5.2 Largest reported vs recalculated p-value gaps

```
row_by_delta <- row_level[order(-row_level$abs_p_delta, na.last = TRUE), ]
knitr::kable(
  head(row_by_delta[, c("variable", "level", "reported_p", "row_chisq_p", "abs_p_delta")], 10),
  digits = 5
)
```

| variable  | level                                               | re-ported_p | row_chisq | pabs_p_delta |
|-----------|-----------------------------------------------------|-------------|-----------|--------------|
| Ethnicity | White                                               | NA          | 0.46069   | NA           |
| Ethnicity | White English/Welsh/Scottish/Northern Irish/British | NA          | 0.64099   | NA           |
| Ethnicity | White Irish                                         | NA          | 0.30610   | NA           |
| Ethnicity | Gypsy or Irish Traveller                            | NA          | 0.31522   | NA           |
| Ethnicity | Any other White background                          | NA          | 0.96745   | NA           |
| Ethnicity | Mixed                                               | NA          | 0.98032   | NA           |
| Ethnicity | White and Black Caribbean                           | NA          | 0.47179   | NA           |
| Ethnicity | White and Black African                             | NA          | 0.71359   | NA           |
| Ethnicity | White and Asian                                     | NA          | 0.66140   | NA           |
| Ethnicity | Any other mixed/multiple ethnic background          | NA          | 0.98739   | NA           |

## 6 simdistr Variable-Level Signals

```
sim_by_p <- sim_var[order(sim_var$simdistr_pvalue, na.last = TRUE), ]
knitr::kable(head(sim_by_p[, c("variable", "level", "simdistr_pvalue")], 15), digits = 5)
```

|    | variable      | level                                       | simdistr_pvalue |
|----|---------------|---------------------------------------------|-----------------|
| 37 | Comorbidities | Solid Tumour Malignancy                     | 0.0543          |
| 30 | Comorbidities | COPD                                        | 0.1052          |
| 43 | Comorbidities | Haematological malignancy                   | 0.1199          |
| 5  | Ethnicity     | Any other White background                  | 0.1218          |
| 42 | Comorbidities | Learning Disability                         | 0.1235          |
| 32 | Comorbidities | Moderate to severe CKD (Grade III or above) | 0.1266          |
| 6  | Ethnicity     | Mixed                                       | 0.1865          |
| 18 | Ethnicity     | African                                     | 0.2060          |

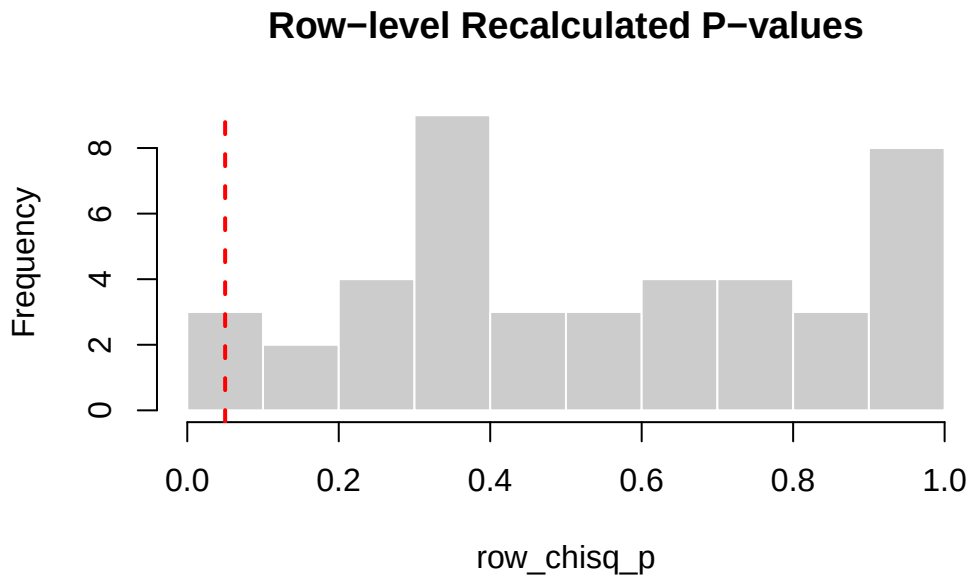


|    | variable                       | level                                               | simdistr_pvalue |
|----|--------------------------------|-----------------------------------------------------|-----------------|
| 20 | Ethnicity                      | Any other Black/African/Caribbean background        | 0.2486          |
| 26 | Number of Listed Comorbidities | 1                                                   | 0.2518          |
| 36 | Comorbidities                  | Physical Disability                                 | 0.2560          |
| 10 | Ethnicity                      | Any other mixed/multiple ethnic background          | 0.2874          |
| 24 | Ethnicity                      | Missing/not declared                                | 0.3278          |
| 2  | Ethnicity                      | White English/Welsh/Scottish/Northern Irish/British | 0.3404          |
| 15 | Ethnicity                      | Chinese                                             | 0.3430          |

## 7 Diagnostics

### 7.1 Distribution of recalculated row p-values

```
hist(
  row_level$row_chisq_p,
  breaks = 10,
  col = "gray80",
  border = "white",
  main = "Row-level Recalculated P-values",
  xlab = "row_chisq_p"
)
abline(v = 0.05, col = "red", lwd = 2, lty = 2)
```



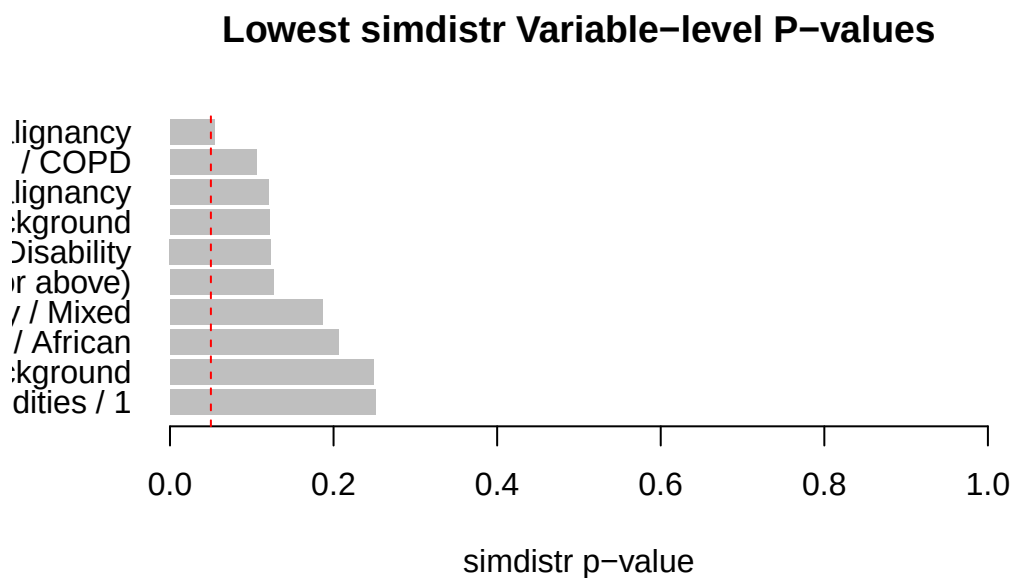
## 7.2 Reported vs recalculated row p-values

```
plot_rows <- row_level[is.finite(row_level$reported_p) & is.finite(row_level$row_chisq_p), ]
if (nrow(plot_rows) > 0) {
  plot(
    plot_rows$reported_p,
    plot_rows$row_chisq_p,
    pch = 19,
    col = "steelblue",
    xlab = "Reported p-value",
    ylab = "Recalculated row p-value",
    main = "Reported vs Recalculated Baseline P-values"
  )
  abline(0, 1, col = "darkgray", lwd = 2)
  abline(h = 0.05, col = "red", lty = 2)
  abline(v = 0.05, col = "red", lty = 2)
} else {
  cat("No rows contain both reported and recalculated p-values; scatter plot skipped.")
}
```

No rows contain both reported and recalculated p-values; scatter plot skipped.

### 7.3 Lowest simdistr variable-level p-values

```
plot_df <- head(sim_by_p, 10)
barplot(
  height = rev(plot_df$simdistr_pvalue),
  names.arg = rev(paste(plot_df$variable, plot_df$level, sep = " / ")),
  horiz = TRUE,
  las = 1,
  col = "gray75",
  border = NA,
  main = "Lowest simdistr Variable-level P-values",
  xlab = "simdistr p-value",
  xlim = c(0, 1)
)
abline(v = 0.05, col = "red", lty = 2)
```



## 8 Interpretation Notes

- This is a baseline-balance forensic screen, not a causal efficacy analysis.
- Row-level p-values depend on one-vs-rest encodings for multi-level variables.

- Reported manuscript p-values may use test choices that differ from recalculation methods.
- Joint p-value methods are sensitive to correlation among baseline variables.